NIKIL KUMAR

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Senior Backend Engineer | Python · FastAPI · Odoo ERP · Laravel · AWS

Senior Backend Engineer with 5 years building production-grade APIs,
ERP systems, real-time platforms, and multi-tenant SaaS products.
Specialises in Python, FastAPI, Odoo ERP (16/17/18/19), Laravel 11,
and AWS. Available immediately for remote roles worldwide.
Key Achievements:
- Architected 117-table HRMS SaaS migration — monolithic Laravel to
dedicated-database-per-tenant architecture with zero data loss
- Built FastAPI backend for live social media app — WebSocket real-time
chat, Redis Pub/Sub, JWT auth, Firebase FCM, PostgreSQL, AWS EC2
- Delivered REST APIs for Flutter mobile apps across transport, food
delivery, healthcare, and ERP domains — serving real production users
- Odoo ERP specialist across 4 versions (16/17/18/19) — custom modules,
Python ORM, QWeb reports, portal integration, view inheritance
- Integrated Razorpay, MSG91, Google Maps API, WhatsApp Business API,
and Petpooja POS in production systems
AWS Cloud Practitioner — In Progress (Target: Q3 2026).

WORK EXPERIENCE

Petrocon • Mangalore, India **12/2024 - Present**
Senior Backend Engineer

- Architected HRMS multi-tenant SaaS — migrated 117-table monolithic Laravel app to dedicated-database-per-tenant using stancl/tenancy, TenantMiddleware, and subdomain routing with zero data loss and complete cross-tenant isolation.
- Built FastAPI backend for Gucchui social media app — async REST APIs, WebSocket real-time chat with Redis Pub/Sub, JWT authentication, SQLAlchemy async ORM, Cloudinary media storage, and Firebase FCM push notifications.
- Designed RESTful APIs in Laravel 11 consumed by Flutter mobile apps — booking lifecycle, OTP-verified delivery, real-time GPS tracking via WebSockets (Laravel Reverb) at 5-second update intervals.
- Built Odoo 18/19 ERP modules for RSL enterprise project — custom Python models, _inherit view inheritance, ORM queries, QWeb PDF reports, and record-level security rules.
- Integrated third-party services at scale: Razorpay (payments + webhooks), MSG91 (OTP/SMS), Google Maps API (geocoding, routing, Haversine), Petpooja POS (menu sync), WhatsApp Business API.

- Deployed full AWS stack — EC2, RDS MySQL, S3, IAM, Nginx, Cloudflare CDN, Supervisor queue workers — maintaining zero-downtime production systems.
- Implemented async job processing with Laravel Queues, Redis, and Jobs — decoupled notifications, SMS, and confirmations from API cycle, eliminating blocking under concurrent load.

Invenger Cloud • Mangalore, India **12/2023 - 11/2024**
Software Engineer

- Built complete Dental Clinic Management module in Odoo 16/17 — appointments, prescriptions, billing, QWeb PDF report templates, and Odoo Studio customisation.
- Integrated WhatsApp Business API for automated patient reminders via Odoo messaging framework — reduced no-show rates.
- Implemented XML view inheritance across 6 existing Odoo modules — extended functionality without modifying core, maintaining upgrade safety from v16 to v17.

Invenger Technologies Pvt. Ltd. • Mangalore, India **08/2021 - 11/2023**
Software Engineer

- Built and maintained Claims Pay — high-volume insurance claims backend in PHP with OOP, MVC, SOLID principles, and PHPUnit test coverage.
- Optimised complex SQL queries for financial reporting — improved performance using indexes and EXPLAIN ANALYZE on large datasets.

Udupi Web Solution • Udupi, India **02/2021 - 08/2021**
Software Engineer

- Built invoice generation system in PHP and Laravel — GST calculation, discount rules, client-specific billing logic.

EDUCATION

B.E. Electronics and Communication Engineering
Shri Madhwa Vadiraja Institute of Technology and Management • Udupi 01/2016 - 01/2020

CERTIFICATIONS

AWS Certified Cloud Practitioner

GitHub Copilot — Active daily practitioner

PROJECTS

Feedygo — Food Delivery Platform

- Petpooja POS integration — real-time menu sync and order management
- MSG91 OTP authentication for customers and delivery partners
- Flutter REST API — real-time order tracking and status updates
- AWS deployment with auto-scaling queue workers

- Stack: Laravel 11 | Petpooja POS | MSG91 | AWS | MySQL | Redis

Dental Clinic Management — Odoo 16/17

- Complete Odoo Python module: appointments, prescriptions, billing
- WhatsApp Business API — automated patient reminders
- QWeb PDF reports | XML view inheritance | Odoo Studio
- Zero-downtime AWS deployments
- Stack: Odoo 16/17 | Python | XML | PostgreSQL | AWS

Furniture Vendor System — Odoo 19

- 4-role workflow: Customer, Supervisor, Vendor, Admin
- 12-state order lifecycle with advance + final invoice automation
- Customer portal | production tracking | QWeb PDF reports
- Role-based access control with record-level security
- Stack: Odoo 19| Python | XML | PostgreSQL | AWS EC2

HRMS — Multi-Tenant SaaS Migration

- Monolithic Laravel to dedicated-DB-per-tenant SaaS architecture
- stancl/tenancy | subdomain routing | TenantMiddleware
- Attendance, Leave, Payroll, IT Tickets, Employee Portal
- 117-table schema migrated with zero data loss
- Stack: Laravel 11 | MySQL | AWS | stancl/tenancy | REST API

LevityX — General Store POS System

- Odoo Community POS + custom credit (Udhaar) tracking module
- WhatsApp bill delivery replacing thermal printer
- Digital customer database with purchase history
- Planned: Flutter customer app + delivery module
- Stack: Odoo 19 | Python | PostgreSQL | WhatsApp API

Gucchui — Social Media App Backend — Freelance Backend Engineer (Paid)

- Built complete REST API backend for Flutter social media app — auth, profiles, friends, posts, stories, real-time chat, notifications.
- WebSocket real-time chat with Redis Pub/Sub — supports multiple concurrent server instances without message loss.
- JWT + bcrypt authentication, Firebase FCM push notifications (background + killed app), Cloudinary media storage.
- Deployed on AWS EC2 with Nginx, SSL (Let's Encrypt), environment-isolated Docker-adjacent Linux setup.

Darmadar — Animal Transport Platform — Senior Backend Engineer

- Three-sided REST API marketplace — Customer App, Driver App, Admin Panel — consumed by Flutter mobile apps.

- Real-time GPS WebSockets (Laravel Reverb) — 5-second live driver location broadcast with sub-second delivery.
- Razorpay payment gateway + webhooks, MSG91 OTP, Google Maps API, Haversine formula driver assignment.

Laundry Management System — Odoo 18 — Odoo Developer

- Complete Odoo 18 module — order workflow, worker assignment, washing types, additional services, invoice generation.
- QWeb PDF reports, smart buttons, state machine (draft → confirmed → in_progress → done → cancelled), mail.thread chatter.
- Integrated Odoo Accounting for advance and final payment automation.

SKILLS

- **Core Language:** Python 3.11+ — OOP, decorators, async/await, type hints, generators, context managers
- **Odoo ERP:** cron, _inherit, Odoo 16/17/18/19 — custom modules, Odoo 16/17/18/19 — custom modules, Python ORM, QWeb, XML views, _inherit, wizards, cron jobs, security rules, portal controllers, XMLRPC, ORM, portal, QWeb, security, wizards, XMLRPC, XML views
- **Backend / API:** API versioning, FastAPI (learning — SQLAlchemy + Pydantic + JWT), Laravel 11, PHP 8.2 (OOP, MVC, SOLID), REST API design, Webhooks
- **Integrations:** Cloudinary, FCM Push, Firebase/FCM, Google Maps API, Moyasar Payment, MSG91, MSG91 (OTP/SMS), Petpooja POS, Razorpay, WhatsApp Business API, XMLRPC
- **Cloud / AWS:** Cloudflare, Cloudflare CDN, CloudFront — AWS Cloud Practitioner in progress, EC2, IAM, Nginx, RDS, S3, SES, Ubuntu server
- **Database:** MySQL 8, PostgreSQL (indexing, EXPLAIN ANALYZE, transactions), PostgreSQL (indexing, EXPLAIN ANALYZE, transactions, query optimisation), Redis
- **Real-time:** Celery, Celery (basics), event broadcasting, Laravel Queues, Laravel Queues and Jobs, Redis Pub/Sub, WebSockets (FastAPI + Laravel Reverb), WebSockets (Laravel Reverb)
- **DevOps:** Agile/Scrum, Docker, Git, GitHub, GitHub Actions (CI/CD), GitHub Actions CI/CD, GitLab, Jira, Linux/Ubuntu, Supervisor
- **AI Tools:** Claude AI — architecture planning, cursor, GitHub Copilot (daily — prompt engineering, code review), GitHub Copilot (daily — prompt engineering, code review, standards)
- **Auth / Security:** bcrypt, HTTPS/SSL, JWT, Laravel Sanctum, Odoo record rules, OTP auth, OTP authentication, RBAC, Record rules (Odoo)
- **Python / FastAPI:** Celery, FastAPI, Pydantic v2, Python 3.11+ (OOP, async/await, decorators, type hints), SQLAlchemy async
- **Laravel / PHP:** Jobs, Laravel 11, PHP 8.2 (OOP, MVC, SOLID), Queues, REST API design, Reverb WebSockets, Sanctum